# FLEET TEAM DIGEST

## 737MAX-FTD-31-24004

**Issue Title :** In-Service Issues with Airplane Condition Monitoring System (ACMS) Block Point (BP) 1.3 Baseline Database (DB)

| Airplane Model | ATA | Minor Model(s) |
|---|---|---|
| 737MAX | 3131-00 | -10, -7, -8, -8200, -9, -BBJ |
| **Other Model(s)** | **ECCN** | |
| | 9E991 | |

| Originated Date | Last Revised Date | Created On |
|---|---|---|
| 04/22/2024 | 11/26/2025 | 10/27/2025 |
| **Estimated Completion Date** | **Next Update Date** | |
| 01/09/2026 | 12/22/2025 | |

## Revision Description

Final Action and Milestones sections were updated to reflect the new ECD for new ACMS BP 1.5 software release, which changed to December 2025.

Production incorporation changed to LN 9600.

## Description

***This issue is only applicable to 31 DFDAU ACMS BASELINE DB P/N: BCG3B-L200-0003.***

737MAX Airplane Condition Monitoring System (ACMS) software update Block Point (BP) 1.3 was released to the in-service fleet in December 2023 via 737-SL-31-084. This release included a new '31 DFDAU ACMS BASELINE DB' P/N: BCG3B-L200-0003, which is a new version of the previous P/N: BCG3A-L200-0002. With this version, several issues have been identified, which are mostly related to new features not functioning as intended or intended fixes that are still not resolved.

**Issues Identified:**

- B07 Report (ECS Under-Overpressure Temperature Manifold High and Low) with updated trigger logic is still over-triggering
  - Boeing updated the B07 (Environmental Controls Pressure and Temperature Manifold Report) Report logic in an attempt to fix a prior ‘over-triggering’ issue, which was documented in 737MAX-FTD-31-20002. The prior issue where the report triggers more often than intended is still occurring with the new software, despite having modified trigger thresholds.
- Most ACMS Reports still do not have any limitation to the amount of reports that can trigger and downlink per-flight
  - Some report retention policies do not limit the number of reports that can trigger per flight leg. This can result in excessive downlinking during certain situations when anomaly reports are triggering.
- New ACMS Backup Flight Phase computation is susceptible to calculating wrong phase during PSEU Air Mode on-ground
  - When the airplane is put into ‘Air Mode’ using the PSEU special function, ACMS will invalidate the OMF provided flight phase, reverting to the backup calculation. This ACMS-computed flight phase will then end up in an in-air phase, typically Climb/Cruise/Flare. This causes the B36/B37 Engine Fail reports to trigger due to Fuel Cutoff while “in-air”. Additionally, B06 Emergency reports will trigger due to Airspeed < 100 Kts while “in-air” and this will trigger every 10 seconds while the

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airplane is in PSEU air-mode due to the previous pre-flight trigger limit issue.

- New CFM-requested Main Transfer Bus Power Available (MTBPA) Timer not calculating correctly
  - The new MTBPA timer value is only being recorded in the first engine start report and is zero in the second engine start report. Additionally, the single MTBPA timer value has been found to be much lower than expected (typically less than one minute) for most flight legs. Root cause investigation found that the timer functions correctly if there is a power cycle between engine shutdown and subsequent engine start. If the airplane has constant power, then the timer resets and continues counting at that the next engine start. The engine start report has a short data collection period that occurs while the MTBPA timer increases, and then logs that small value (typically less than one minute). Additionally, the timer is being cleared after the first engine start report.
- New EEC Max Climb N1 Ratings for the right engine in the Climb Reports (for all 5 Altitude Climb levels) are always being recorded as zero
  - This was a new parameter addition in the climb reports and it is always zero for the right engine in the climb reports due to incorrect A429 SDI definition in the ACMS database.
- B31 Climb Report is over-counting in the B42 Flight Summary Report
  - Previously the B31 report would be counted multiple times in the flight summary report, despite only triggering one time - due to an issue with a logic unit erroneously increasing the report count at several flight phase transitions.
- New 'ECS Units' parameter is always invalid in B07/B08/B47/B48 reports
  - This parameter was added to provide reference to which units (Fahrenheit or Celsius) are selected, however it has been found to be the invalid character in all reports. Root cause was found to be the parameter definition using A429 BNR (binary) type SSM instead of DIS (discrete) type SSM resulting in the signal always being flagged as 'Failure Warning'.

## Status

Boeing has been working with several operators to obtain in-service data, which has been instrumental in determining root cause for these issues. Root cause has been identified for each of the issues and corrective action has been identified for all issues except the B07 report triggering.

A study is being performed to categorize the various trigger reasons seen in service for the B07 reports being generated to adequately adjust the trigger thresholds.

Most of the planned changes & issue resolutions are small adjustments to parameter definitions or logic units. The new ACMS flight phase logic will be a substantial update and the new MTBPA timer will have a significant logic adjustment to resolve these issues for enhanced robustness (including unique counting for each engine).

There are differences between the OMF Flight Phase calculation, the new ACMS Flight Phase calculation and the ACMS "validation" of OMF phase is not robust enough. When the airplane is put into 'air mode' (via PSEU special function), the OMF remains in power-on phase. The ACMS sees air/gnd = AIR, OMF phase – PO and marks it as invalid, reverting to it's own calculation. ACMS has less logic versus OMF (not looking for engines on or ground speed for some in-air phases) and that results in it calculating some in-air phase. Boeing will be adjusting the ACMS flight phase calculations to better match the OMF logic and increase the reliability of the 'redundant' calculation.

### New Issue Discovered May 2024 ###

*This issue applies to ACMS BASELINE DB P/Ns: BCG3A-L200-0002 and BCG3B-L200-0003*

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Operators recently reported a high volume of B09 (Temperature Control Valve (TCV) and Trim Air Valve (TAV) Limit Switch) Reports triggering which led Boeing to initiate an investigation. The B09 reports identified were all triggered due to simultaneous limit switch alerts, however when analyzing associated ADR-ALL data, the related open/closed limit switches were not active simultaneously. Boeing found that the B09 report logic unit has a flag for when the first limit switch goes active, but that flag does not clear/reset after the switch changes to inactive. If at a later time the other limit switch goes active, ACMS will erroneously trigger the B09 report even though both limit switches are not active simultaneously. At this time, it is advised to disable the B09 report generation.

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Status as of November 2025:

The root cause investigation for all issues discussed above is complete and development of the new ACMS Baseline DB is currently in-work.

## Interim Action

Boeing advises operators to disable the new ACMS backup Flight Phase calculation and to disable the B07/B47 ACMS reports using the following procedure:

**Disable the ACMS Backup Flight Phase calculation.**

*This can be accomplished on wing or via uplink command.

**On-Wing:**

AMM TASK 31-31-22-970-803 to change modifiable constants.

1. Using the MCDU, navigate to ACMS  REPROGRAM  MODIFY CONSTANTS  CHANGE CONSTANTS
2. Enter ‘FPBACKUP/B’ on the scratchpad and select L1
3. A ‘1’ should appear on the DEF VALUE line
4. Enter ‘0’ on the scratchpad and select L2
5. A ‘0’ should appear on the MOD VALUE line.
6. Select L5 to save the modified setting.

**ACARS Uplink:**

Send this uplink request to the DFDAU (identifier = DFD)

- ‘XCM B FPBACKUP 0’

**Disable the B07, B09, and B47 Report generation.**

*These reports have ACARS auto-downlink disabled by default, but the generation is enabled by default. This adjustment can be accomplished on wing or via uplink command.

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**On-Wing:**

AMM TASK 31-31-22-860-804 to disable report generation

**ACARS Uplink:**

Send this uplink request to the DFDAU (identifier = DFD)

- 'XRD 0 B07 D'
- 'XRD 0 B09 D'
- 'XRD 0 B47 D'

**Current Mitigation for AHM customers:**

- AHM monitoring for B07/B09/B47 report downlinks and subsequently sending uplink commands to disable the report generation/auto-downlinking

- AHM checking all B99 configuration reports for B07/B09/B47 settings (generate and downlink) and disabling if found to be enabled

- AHM checking all B99 configuration reports for FPBACKUP = 1 and disabling found to be enabled

- AHM monitoring for B36A/B37A (Engine Fail) reports and subsequently sends uplink commands to disable backup flight phase calculation

## Final Action

Boeing is prioritizing development of a new ACMS Baseline DB part to resolve the issues discovered after the recent release of ACMS Baseline DB P/N: BCG3B-L200-0003 and incorporate an updated B21 Landing Report. A Service Letter will be released to announce the new software part availability. The estimated completion date for the BP 1.5 ACMS Baseline DB release is December 2025.

## Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Root Cause Established: April 2024 (Complete)
ACMS BP 1.5 Software Release: December 2025 (Target)
Service Letter Available: December 2025 (Target)

Production Incorporation: LN 9600 | March 2026 (Target)

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## Reference Categories

| Reference Type | Reference Number | Revision Number | Reference Date |
|---|---|---|---|
| Other | BCG3B-L200-0003 | | |
| Service Letter (SL) | 737-SL-31-084 | | |
| Other | BCG3A-L200-0002 | | |

## Related Categories

- Dispatch Reliability Issues : Issues that cause revenue flight delays and cancellations

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